

# Intro to Materials by Design

## Local Python Environment Setup

This environment is intended for work performed on your personal computer.

It works with:

- macOS
- Windows using WSL and an Ubuntu terminal

This environment does **not** require GPU support. A separate environment will be used later for HPC and GPU calculations.

The environment contains packages for:

- Basic scientific Python
- Data analysis
- Plotting
- Machine learning
- Materials science
- Jupyter notebooks

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## 1. Install Miniforge

We will use **Miniforge**, which provides the `conda` environment manager and uses the `conda-forge` package repository.

### macOS

Open the normal **Terminal** application.

### Windows

Open your **Ubuntu terminal through WSL**.

Do not run these commands in Windows PowerShell or Command Prompt. For this course, the Python environment should live inside the Ubuntu/WSL environment.

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## Check that `curl` is available

Run:

```
curl --version
```

If this prints information about `curl`, continue to the next step.

On Ubuntu/WSL, if `curl` is not installed, run:

```
sudo apt update  
sudo apt install -y curl
```

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## Download Miniforge

The following command automatically selects the correct installer for macOS or Linux and for the computer's processor architecture.

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/  
Miniforge3-$(uname)-$(uname -m).sh"
```

Run the installer:

```
bash Miniforge3-$(uname)-$(uname -m).sh
```

The installer will ask several questions.

In most cases, you can accept the default options.

When asked whether Miniforge should initialize conda for your shell, answer:

```
yes
```

When installation is finished, close the terminal and open a new one.

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## 2. Check that conda works

Run:

```
conda --version
```

You should see something similar to:

```
conda 26.x.x
```

The exact version number may be different.

If you see a version number, conda is working.

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## 3. Set conda-forge to strict channel priority

Run:

```
conda config --set channel_priority strict
```

This helps conda consistently obtain compatible packages from the same software ecosystem.

You only need to do this once.

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## 4. Download the course environment file

Place the provided file

```
environment_local.yml
```

in a convenient directory.

For example, you may make a directory for the course:

```
mkdir materials_by_design  
cd materials_by_design
```

Then place `environment_local.yml` in this directory.

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## 5. Create the course environment

Make sure that your terminal is in the same directory as `environment_local.yml`.

You can check the files in the current directory with:

```
ls
```

You should see:

```
environment_local.yml
```

Now create the environment:

```
conda env create -f environment_local.yml
```

Conda will download and install the required packages.

The environment is named:

```
materials_by_design_local
```

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## 6. Activate the environment

Run:

```
conda activate materials_by_design_local
```

Your terminal should now show the environment name near the beginning of the command line.

For example:

```
(materials_by_design_local) student@computer:~$
```

This indicates that Python commands will use the course environment.

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## 7. Install PyTorch

PyTorch is installed after creating the conda environment.

The command is slightly different on macOS and Ubuntu/WSL.

### macOS

Run:

```
python -m pip install torch
```

### Windows using Ubuntu/WSL

Run:

```
python -m pip install torch --index-url https://download.pytorch.org/whl/cpu
```

The WSL command explicitly installs the CPU version of PyTorch.

GPU support is not required for the local course environment.

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## 8. Check the Python version

Run:

```
python --version
```

You should see Python 3.12.

For example:

```
Python 3.12.x
```

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## 9. Test the environment

Place the provided file

```
verify_local_environment.py
```

in your working directory.

Run:

```
python verify_local_environment.py
```

The script tests the major packages used in the course.

If everything is installed correctly, the final line should say:

```
Environment check completed successfully.
```

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## 10. Packages included in the environment

The environment contains the following major packages.

### Scientific computing

```
numpy  
scipy
```

### Data analysis

```
pandas  
openpyxl  
h5py
```

## Plotting

```
matplotlib
```

## Machine learning

```
scikit-learn  
torch
```

## Materials science

```
ase  
pymatgen
```

## Additional utilities

```
networkx  
pyyaml  
tqdm
```

## Interactive Python

```
jupyterlab  
ipykernel
```

Additional packages may be added later in the course.

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# 11. Running the introductory Python examples

First activate the environment:

```
conda activate materials_by_design_local
```

Move into the directory containing the Python scripts.

For example:

```
cd python_basics
```

Run a script with:

```
python 00_python_basics_overview.py
```

Then continue through the examples:

```
python 01_logic_and_loops.py  
python 02_lists_dictionaries_and_functions.py
```

and so on.

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## 12. Starting JupyterLab

Activate the environment:

```
conda activate materials_by_design_local
```

Then run:

```
jupyter lab
```

JupyterLab should open in your web browser.

The terminal running JupyterLab should remain open while you are using it.

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## 13. Leaving the environment

When you are finished working, run:

```
conda deactivate
```

The environment name should disappear from the beginning of the terminal prompt.

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## 14. Using the environment again later

You do **not** need to reinstall the environment every time you work on the course.

Simply open a terminal and run:

```
conda activate materials_by_design_local
```

Then move to your course directory and run your scripts.

For example:

```
cd materials_by_design  
python example.py
```

When finished:

```
conda deactivate
```

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## 15. Checking installed packages

To see all packages installed in the environment:

```
conda list
```

To check whether a particular package is installed:

```
conda list numpy
```

or:

```
conda list pymatgen
```

For packages installed using pip, you can also use:

```
python -m pip list
```

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## 16. If something goes wrong

First make sure the correct environment is activated:

```
conda activate materials_by_design_local
```

Then check which Python executable is being used:

```
which python
```

On macOS or Ubuntu/WSL, the path should contain something similar to:

```
miniforge3/envs/materials_by_design_local
```

You can also rerun:

```
python verify_local_environment.py
```

and look for the first package that produces an error.

Do not install missing course packages into the system Python installation. Install them only after activating the course conda environment.